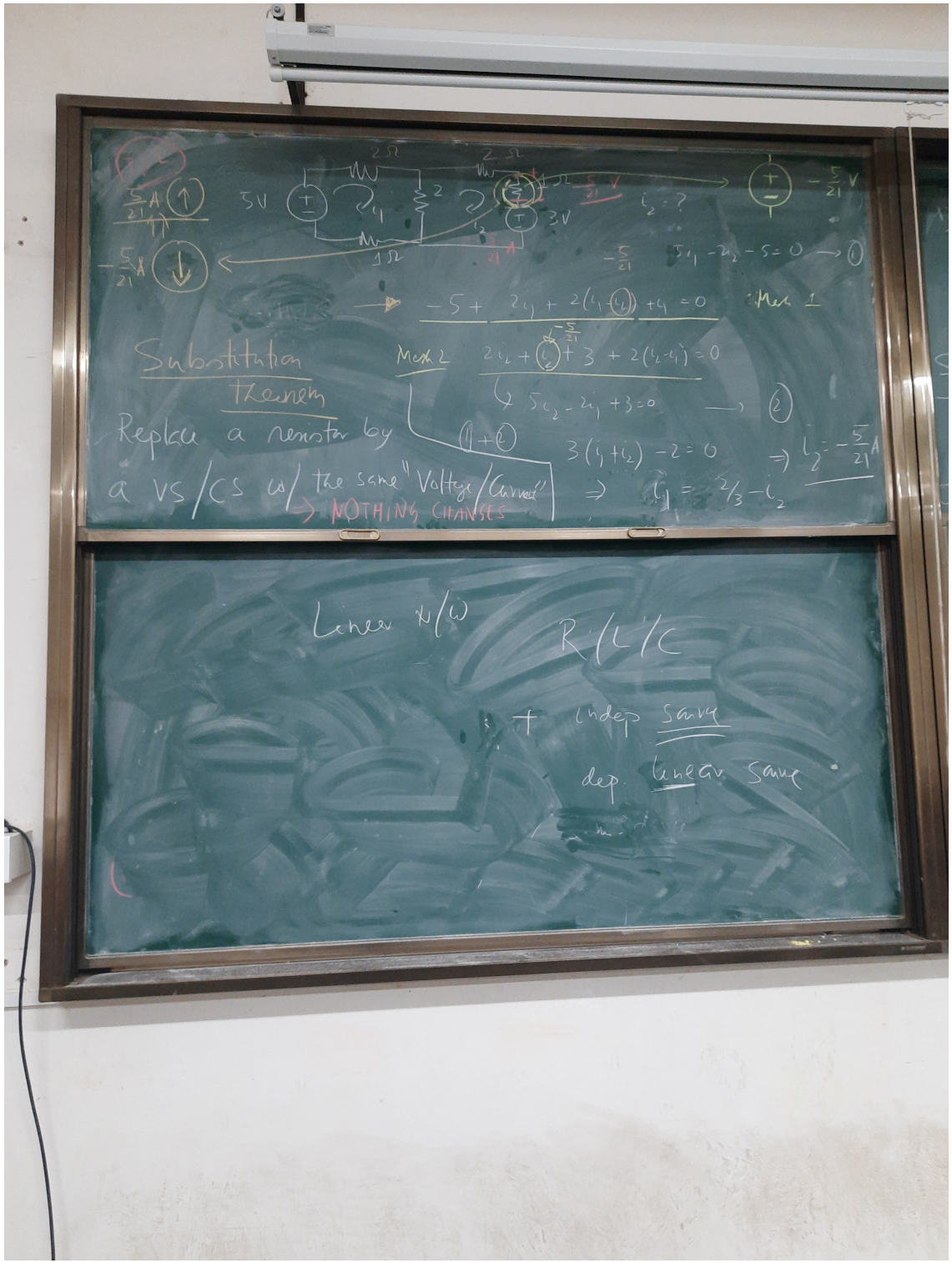
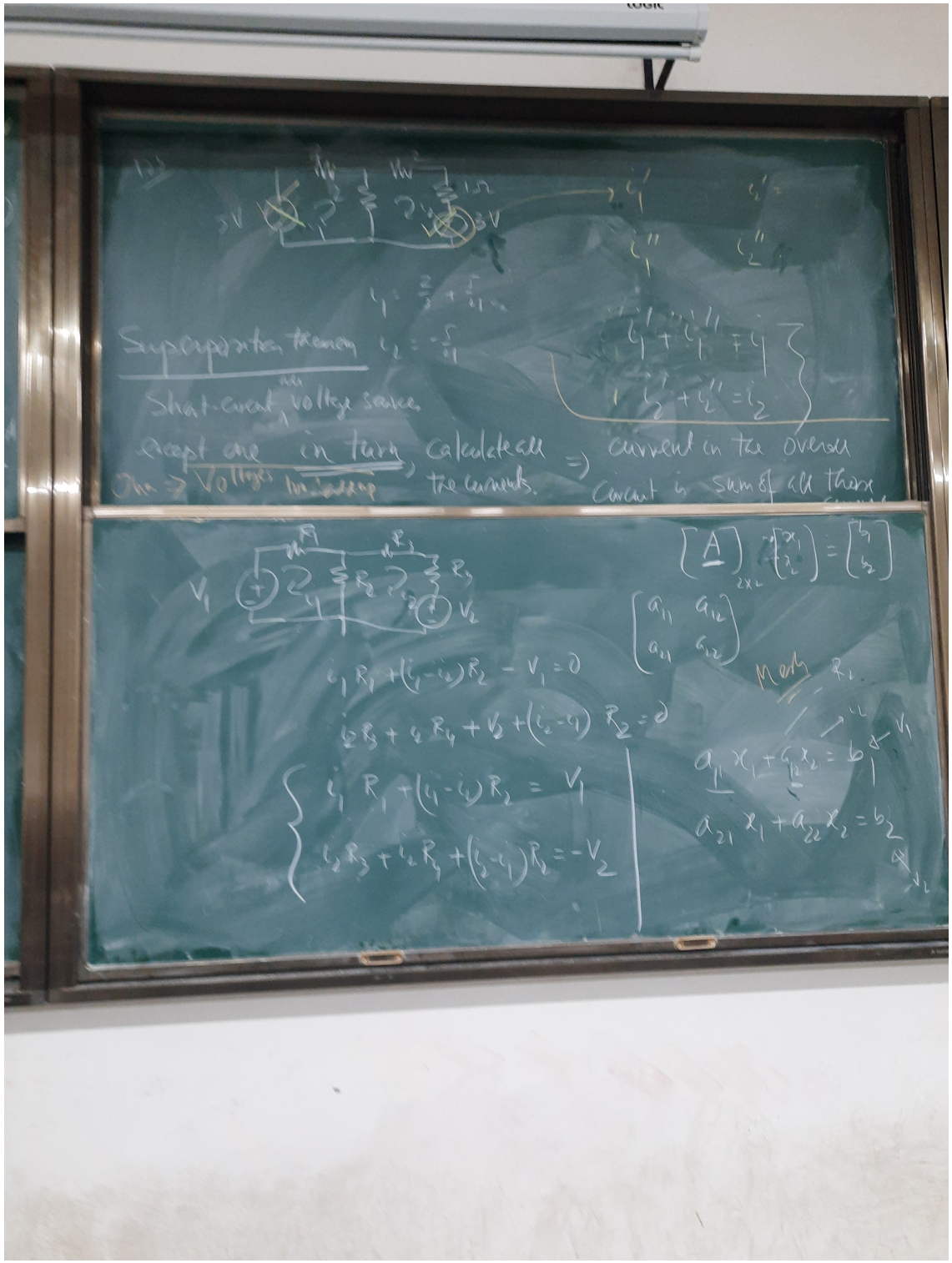
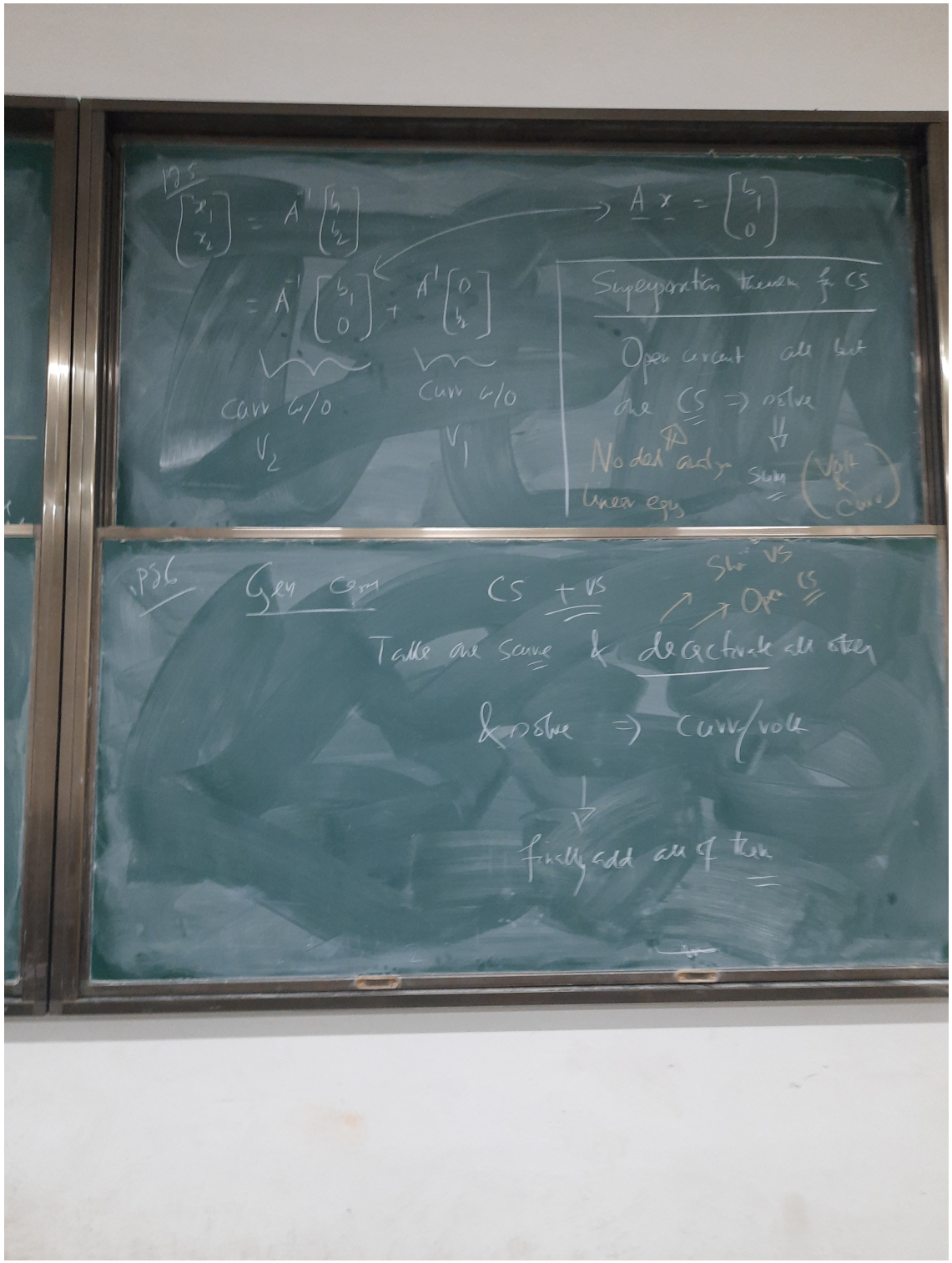

2Ω
2Ω
5V
3V
1Ω
$i_2 = ?$
$5i_1 - 2i_2 - 5 = 0$ → ①
$-5 + 2i_1 + 2(i_1 - i_2) + 4 = 0$ Mesh 1
Mesh 2
$2i_2 + 3 + 2(i_2 - i_1) = 0$
$5i_2 - 2i_1 + 3 = 0$ → ②
① + ②
$3(i_1 + i_2) - 2 = 0$ ⇒ $i_2 = -\frac{5}{21}A$
⇒ $i_1 = \frac{2}{3} - i_2$
Substitution theorem
Replace a resistor by a VS/CS w/ the same "Voltage/Current"
→ NOTHING CHANGES
Linear N/W
R/L/C
+ indep sources
dep linear sources


Superposition theorem
Short-circuit all voltage source except one in turn, calculate all the currents.
Then ⇒ Voltages
⇒ current in the overall circuit is sum of all these
Mesh


Superposition theorem for CS
Open circuit all but one CS
Nodal analysis
Linear eqns
Sum
Volt
Curr
Grey Com
CS ± VS
Short VS
Open CS
Take one source & deactivate all others
& solve ⇒ curr/volt
finally add all of them